
RAWHATUR RABBI

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Sub-Project Manager

Office of the HEAT Sub-Project

Department of Computer Science and Engineering

Green University of Bangladesh

Purbachal American City, Kanchan, Rupganj, Narayanganj-1461

Subject: Application for Research Assistant Position - HEAT Sub-Project (BDREN A-Z)

Dear Sir/Madam,

I am writing to express my enthusiastic interest in the Research Assistant position for the HEAT Sub-Project at Green University, as advertised. As a current Research Assistant at BRAC University with a strong background in deep learning and hands-on experience building Large Language Model (LLM) applications, I am thrilled by the opportunity to contribute to the "BDREN A-Z" project.

The position's requirements are a precise match for my qualifications. I am currently pursuing my MSc in Computer Science and Engineering at BRAC University, having already completed my BSc. My research primarily focuses on AI for Healthcare, Natural Language Processing, and Human-Centered Machine Learning Systems. I have co-authored several peer-reviewed publications in Nature Scientific Reports and IEEE conferences, including works on representation learning, spiking neural networks, and RAG-based AI systems.

My practical experience has prepared me to contribute immediately. I led the development of an agentic RAG framework for the "AgriBuddy" project using Bangla embeddings. Concurrently, I am collaborating on a Carnegie Mellon University PhD project to develop an automated ICU documentation system, applying VLM principles and few-shot prompting to recognize and summarize clinical activities from video feeds. This combined experience with RAG and applied VLMs is directly applicable to your goal of building an LLM framework for disease detection and solution recommendations. My grant-funded research and publication record further demonstrate my capability to manage research from conception to publication.

I am particularly excited by Green University's effort to create a national research collaboration network utilizing LLMs and graph neural networks, as this aligns perfectly with my current work on multimodal representation learning. Given this strong alignment, I am confident that my hands-on experience with RAG, my deep learning background, and my passion for research make me an ideal candidate. I have attached my CV for your review and am eager to discuss how I can contribute to the success of the HEAT Sub-Project.

Thank you for considering my application. I would be eager to discuss how my experience across various directions of AI research could contribute to your ongoing initiatives.

Sincerely,

Rawhatur Rabbi

Research Assistant

Research Seed Grant Initiative Project (2024-2025)

BRAC University

Rawhatur Rabbi

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Summary — Curious and research-driven professional with a strong foundation in Artificial Intelligence. My work centers on developing efficient and interpretable deep learning systems for real-world problems, with a commitment to scientific rigor and open collaboration. Passionate about lifelong learning and discovery, I aim to pursue advanced research that bridges human-centered AI and computational innovation, contributing meaningfully to both academia and applied science.

Research Interests

AI for Healthcare • Multimodal Learning • Explainable AI • Resource-Constrained and Efficient Deep Learning • Representation Learning and Self-Supervised Objectives • Neuromorphic and Spiking Neural Networks • Real-Time Perception and Activity Recognition • Vision-Language Models (VLMs) • Human-Centered and Trustworthy AI Systems

Research & Applied AI Experience

BRAC University

Sep 2023 – Present

Research Assistant (RA)

RSGI 2024 Grant: Real-Time Fall Detection and Activity Recognition to Assist Elderly People

- Led the end-to-end dataset collection initiative, including designing the data collection pipeline, coordinating participant interview sessions, and ensuring ethical and technical compliance during acquisition.
- Developed and deployed a real-time human activity recognition pipeline integrating deep learning and computer vision, achieving 90% classification accuracy on custom motion datasets.
- Engineered internal network architecture to support distributed data processing and real-time streaming for model inference.

School of Data and Sciences, Department of Computer Science and Engineering

- Mentored 30+ undergraduate thesis students, providing end-to-end guidance from research design to publication in peer-reviewed conferences and journals.
- Authored and co-authored multiple scholarly articles accepted in high-impact international venues, contributing to the university's research visibility.
- Developed and optimized neural network architectures for real-time image and video analysis, leveraging transfer learning to achieve 15–20% accuracy gains on domain-specific datasets.
- Contributed to grant proposals and helped secure funding for machine learning and computer vision projects, emphasizing innovation and research scalability.

Brac Business School

- Designed and optimized a relational database for researcher and student records, improving data accessibility and query performance by 40%.
- Conducted data-driven analyses using Python, SQL, and ML toolkits to support ongoing computer vision research.

BRAC University

Jan 2022 – Apr 2022

Undergraduate Teaching Assistant (Student Tutor)

- Provided personalized one-on-one instruction to address individual learning needs.
- Simplified complex math and computer science concepts for better comprehension.
- Tracked student progress and tailored tutoring sessions accordingly.
- Provided coding guidance, helping with algorithm design and debugging.
- Created and shared study materials, including practice problems and solutions.

Professional Experience

Medina Tech Ltd.

Apr 2023 – Sept 2023

Software Engineer (AI & ML)

- Enhanced datasets through data augmentation, diversifying data by 30%, and performed in-depth data analysis, leading to a 20% improvement in model performance.
- Implemented geocode visualization methods, identifying location-based trends, and generated reports.
- Collaborated on backend development using Django and Odoo, contributing to the seamless functionality of web applications and databases.

- **Technical Skills:** Python, Keras, Tensorflow, NumPy, Pandas, Scikit-learn, Linux, Nginx, PostgreSQL, Django, Odoo, Git.

Master Wizr

Jul 2021 – Jan 2024

QA & Automation Tester

- Developed and executed comprehensive test plans and cases for software applications to ensure alignment with system specifications.
- Automated repetitive testing tasks using tools like Selenium and Postman, reducing manual testing time by 50%.
- Identified, documented, and reported bugs, tracking them through to resolution to maintain software quality standards.
- **Soft Skills:** Selenium, Pytest, Jira.

Education

BRAC University, Dhaka, Bangladesh

Apr 2024 – Present

MSc in Computer Science and Engineering

- Current CGPA: 3.83/4.00
- Working as a Graduate Research Assistant
- Awarded Merit-Based Scholarship
- **Courses:** Symbolic Machine Learning, Advanced Artificial Intelligence, Information Assurance and Human Factor, Pattern Recognition, Advanced NLP

BRAC University, Dhaka, Bangladesh

Jan 2019 – Dec 2022

BSc in Computer Science and Engineering

- Graduated with high distinction, CGPA: 3.62/4.00
- Worked as a Student Tutor
- **Courses:** Data Structure, Algorithm, Programming for the Internet, Software Engineering, Artificial Intelligence, Machine Learning, Database Design

Notre Dame College, Dhaka, Bangladesh

Mar 2016 – Aug 2018

Higher Secondary School Certificate

- GPA: 5.00/5.00
- Science Stream (Physics, Chemistry, Mathematics, Biology, ICT)

Publications

Published (Journals)

- Datta, J., **Rabbi, R.**, Saha, P. et al. **Deep Representation Learning using Layer-wise VICReg Losses.** *Nature Scientific Reports*, 15, 27049 (2025). DOI: [10.1038/s41598-025-08504-2](https://doi.org/10.1038/s41598-025-08504-2)

Published (Conferences)

- Datta, J., **Rabbi, R.**, Rafin, N. I., and Alam, M. G. R. **Peer-Guided Optimization: Incorporating Collaborative Learning into Stochastic Optimization in Machine Learning.** *2025 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, Chittagong, Bangladesh, pp. 1–6. DOI: [10.1109/ECCE64574.2025.11013057](https://doi.org/10.1109/ECCE64574.2025.11013057)
- Datta, J., Sarwar, F. A., **Rabbi, R.**, Saha, P., Alam, M. G. R., Uddin, J., & Zereen, A. N. (2025). **Parameter-Efficient Image Classification with Convolutional Spiking Neural Network via Fast Sigmoid Surrogate Gradient Descent.** *2025 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, Chittagong, Bangladesh, pp. 1–6. DOI: [10.1109/ECCE64574.2025.11013059](https://doi.org/10.1109/ECCE64574.2025.11013059)
- **Rabbi, R.**, Datta, J., Ahmed, S., & Zereen, A. N. (2024). **Wildfire Prediction using Convolutional Kolmogorov-Arnold Network.** *2024 13th International Conference on Electrical and Computer Engineering (ICECE)*, IEEE, pp. 667–672. DOI: [10.1109/ICECE64886.2024.11024794](https://doi.org/10.1109/ICECE64886.2024.11024794)
- Arefin, M. Y., **Rabbi, R.**, Turna, I. F., Zannat, Z., & Karim, D. Z. (2023). **Lightweight Custom CNNs Take on Corn Leaf Disease: Deep Learning for Plant Pathology.** *2023 3rd International Conference on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME)*, IEEE. DOI: [10.1109/ICECCME57830.2023.10252315](https://doi.org/10.1109/ICECCME57830.2023.10252315)

Under Review

- **SE-CCT: Compact Convolutional Transformer with Channel Attention for Deepfake Image Detection.** Submitted to *Egyptian Informatics Journal*.
- **ACEPnet: An Enhanced CNN-Attention Framework for Predicting National Per Capita Electricity Consumption using Nightlight Satellite Imagery and Geo-Demographic Data.** Submitted to *IEEE Transactions on Geoscience and Remote Sensing*.
- **Optimizing Building Energy Consumption with Genetic Algorithms and Explainable AI.** Submitted to an *IEEE-sponsored international conference*.
- **Low-Light Vision-based Fall Detection: A Proof-of-Concept on Fall Detection** Submitted to an *IEEE-sponsored international conference*.
- **Human Fall Detection Through Velocity-Driven Temporal Deep Learning.** Submitted to an *IEEE-sponsored international conference*.
- **Temporal Fusion of Convolutional and LSTM Networks for Vision-Based Fall Detection Using Anatomical Keypoints.** Submitted to an *IEEE-sponsored international conference*.

Working Manuscripts

- *Minimizing the Effect of Sleep Deprivation in the Forward-Forward Algorithm.* In preparation for submission to *Transactions on Machine Learning Research (TMLR)*.

Projects

Automated ICU Activity Documentation System

PhD Research Project, Carnegie Mellon University

- Collaborating on a PhD research project at Carnegie Mellon University under the supervision of **Dr. Sarah M. Preum**, Assistant Professor, Department of Computer Science, Dartmouth College, and former Postdoctoral Research Scholar at CMU.
- Developing an intelligent ICU documentation system that automatically recognizes and summarizes patient care activities from real-time video feeds.
- Designing a robust *activity recognition pipeline* and experimenting with *few-shot prompting*, *fine-tuning*, and *transfer learning* to improve recognition accuracy in low-data clinical settings.
- Aiming to produce structured, time-stamped documentation to support clinicians and enhance medical record precision.

Video Analytics for Real-Time Fall Detection and Activity Recognition to Assist Elderly People

Research Seed Grant Initiative 2024, BRAC University

- Led the end-to-end development of a fully functional real-time fall detection system using advanced video analytics to enhance elderly safety and independent living.
- Designed and implemented the complete pipeline on *NVIDIA Jetson Orin Nano*, integrating edge-based processing, a user-friendly interface, and cloud deployment for real-time alert delivery to caregivers.
- Curated and annotated a custom dataset of falls and daily activities using pose estimation and machine learning, significantly improving detection accuracy and model robustness.

AgriBuddy

AI Hackathon, Bangladesh Innovation Conclave, BRAC University

- Developed **AgriBuddy**, a multilingual AI-powered agricultural support system providing personalized guidance to Bangladeshi farmers.
- Designed and implemented an agentic Retrieval-Augmented Generation (RAG) framework using Bangla embeddings to deliver context-aware responses via a smart chatbot.
- Authored and publicly released the technical report “AgriBuddy: An AI-powered Agent-based System for Bangladeshi Agriculture,” documenting system architecture, dataset design, evaluation metrics, and deployment strategy.
- Evaluated AgriBuddy on field datasets and demonstrated measurable improvements in recommendation accuracy and user engagement under real farming conditions.

Dr. Chashi

Genius Farms Limited (Old Medina Tech Ltd.)

- Developed and fine-tuned deep learning models for the *Dr. Chashi* digital farming platform using real field-collected datasets, enhancing prediction accuracy and reliability for agricultural insights.
- Executed comprehensive data preprocessing and selection workflows, improving data quality and model performance.
- Collaborated with agricultural experts and researchers to integrate domain knowledge and state-of-the-art innovations, ensuring the platform’s scientific and practical relevance.

Achievements

- Finalist, Agriculture, AI Hackathon, Bangladesh Innovation Conclave (2025)
- Secured Fund of 0.8M BDT, Research Seed Grant Initiative 2024, BRAC University (2024)
- Finalist, Line Follower Robot, RUET CSE FEST, RUET (2022)
- Finalist, Hackathon of Joyjatra'21 National Fest, Bangladesh (2021)
- 2nd Runner-up, Intra-University Programming Contest, BRAC University (2020)
- Vice Chancellor's (VC) list, Listed for outstanding academic performance, BRAC University (2019)
- Life Time Membership, Notre Dame Science Club, NDC (2018)
- 2nd Position, Web Designing Competition, Notre Dame Annual Science Festival 2014 & 24th GKC, NDC (2014)

Skills

Front End:	HTML5, CSS3, JavaScript, React.js
Back End:	Node.js, Express.js, Django, Flask, FastAPI
Database:	MongoDB, MySQL, PostgreSQL, Firebase, ChromaDB, FAISS
Languages:	Python, Java, C++, JavaScript, TypeScript
Mobile App:	Flutter, Java
ML/DL Libraries:	TensorFlow, Keras, PyTorch
Deployment Tools:	Linux, Git, Bash/Zsh, Docker, MLflow, DVC
NLP/LLM:	Transformer-based text models, RAG, Named Entity Recognition, Prompt Engineering, LangChain, LlamaIndex, Hugging Face

Extra Curricular Activities

Instructor, Basic of Robotics Workshop, BRAC University

Jan 2022 – Dec 2022

- Instructed students in Arduino programming, covering coding principles and syntax.
- Demonstrated the practical applications of various sensors, including ultrasonic, infrared, and temperature sensors.
- Guided students in constructing line follower robots, soccer bots, and obstacle-avoidance robots.
- Designed custom circuits for real-life projects, optimizing functionality and efficiency.
- **Technical Skills:** C/C++, Circuit Design, Arduino, Sensor Interfacing.

Director, Robotics Club of BRAC University (ROBU)

Apr 2023 – Sept 2023

- Served as Director of the Robotics Club, overseeing projects, events, and member coordination.
- Led the redesign of ROBU's portfolio website, incorporating modern UI, improved navigation, and enhanced accessibility.
- Organized and managed robotics workshops, seminars, and competitions, fostering engagement among students.
- Prepared and presented financial reports, analyzing budgets and recommending cost-optimization measures.

References

Dr. Md. Golam Rabiul Alam

Professor, Department of Computer Science and Engineering, School of Data and Sciences, BRAC University

Email: rabiul.alam@bracu.ac.bd

Relationship: Project Supervisor

Dr. Swakkhar Shatabda

Professor, Department of Computer Science and Engineering, School of Data and Sciences, BRAC University

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Relationship: Project Supervisor

Dr. Aniqua Nusrat Zereen

Assistant Professor, Department of Computer Science and Engineering, School of Data and Sciences, BRAC University

Lecturer, Faculty of Information and Communication Technology, Mahidol University, Thailand

Email: aniqua@bracu.ac.bd

Relationship: Current Supervisor